COMPUTER ENGINEERING

QUEZON CITY

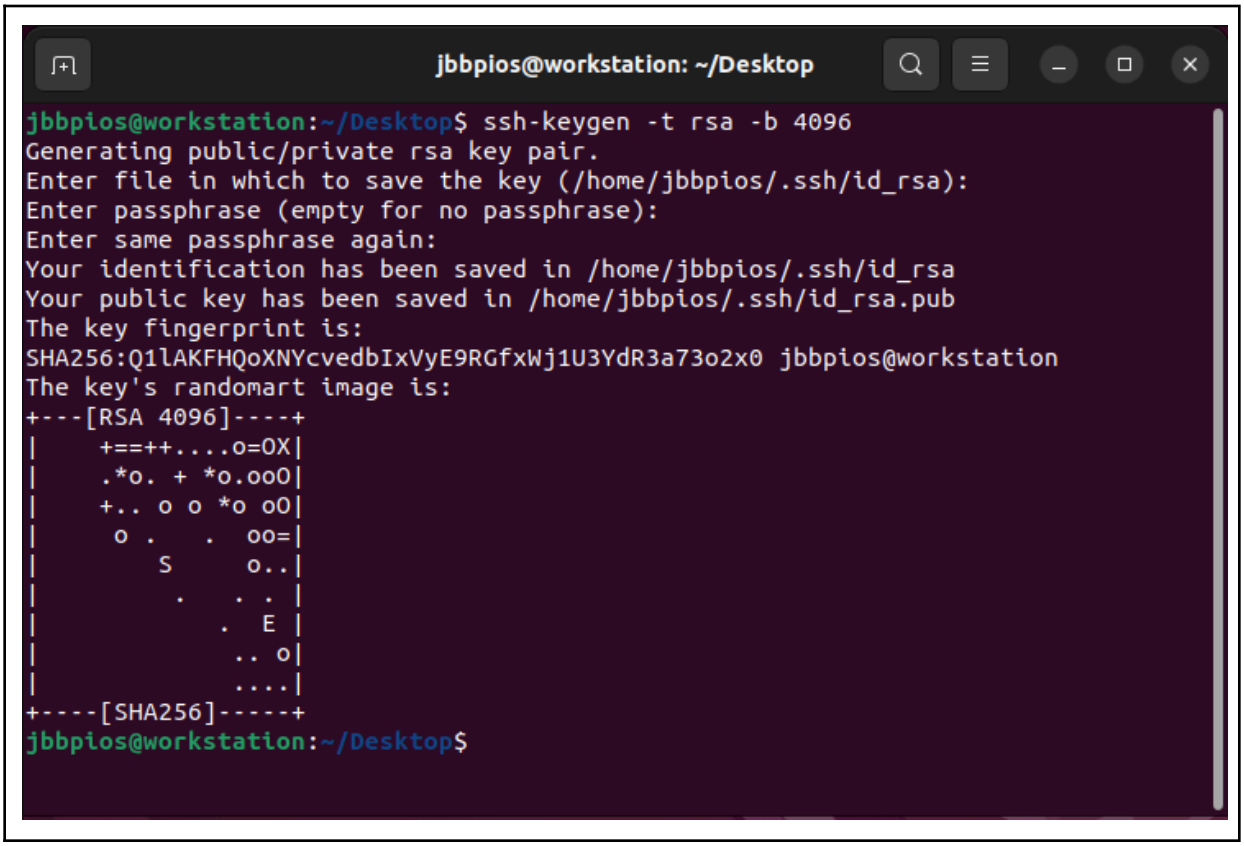
COURSE CODE/TITLE:	CPE 212	**SECTION:**	CPE31S2
DATE PERFORMED:	July 16, 2026	**DATE SUBMITTED:**	July 16, 2026
NAME:	Pios, Joshua Paul B.	**INSTRUCTOR:**	Engr. Jimlord M. Quejado

ACTIVITY NO. 2.1 **SSH Key-Based Authentication and GIT Setup**

1. PROCEDURE OUTPUT

What procedures did you perform?

Task 1



Task 2

```
jbbpios@workstation:~/Desktop$ ssh-copy-id -i ~/.ssh/id_rsa.pub jppios@server1
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/jbbpios/.ssh/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter out any that ar
e already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompted now it is to
 install the new keys
jppios@server1's password:

Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'jppios@server1'"
and check to make sure that only the key(s) you wanted were added.

jbbpios@workstation:~/Desktop$
```

```
jbbpios@workstation:~/Desktop$ ssh-copy-id -i ~/.ssh/id_rsa.pub jpbpios@server2
/usr/bin/ssh-copy-id: INFO: Source of key(s) to be installed: "/home/jbbpios/.ss
h/id_rsa.pub"
/usr/bin/ssh-copy-id: INFO: attempting to log in with the new key(s), to filter
out any that are already installed
/usr/bin/ssh-copy-id: INFO: 1 key(s) remain to be installed -- if you are prompt
ed now it is to install the new keys
jpbpios@server2's password:

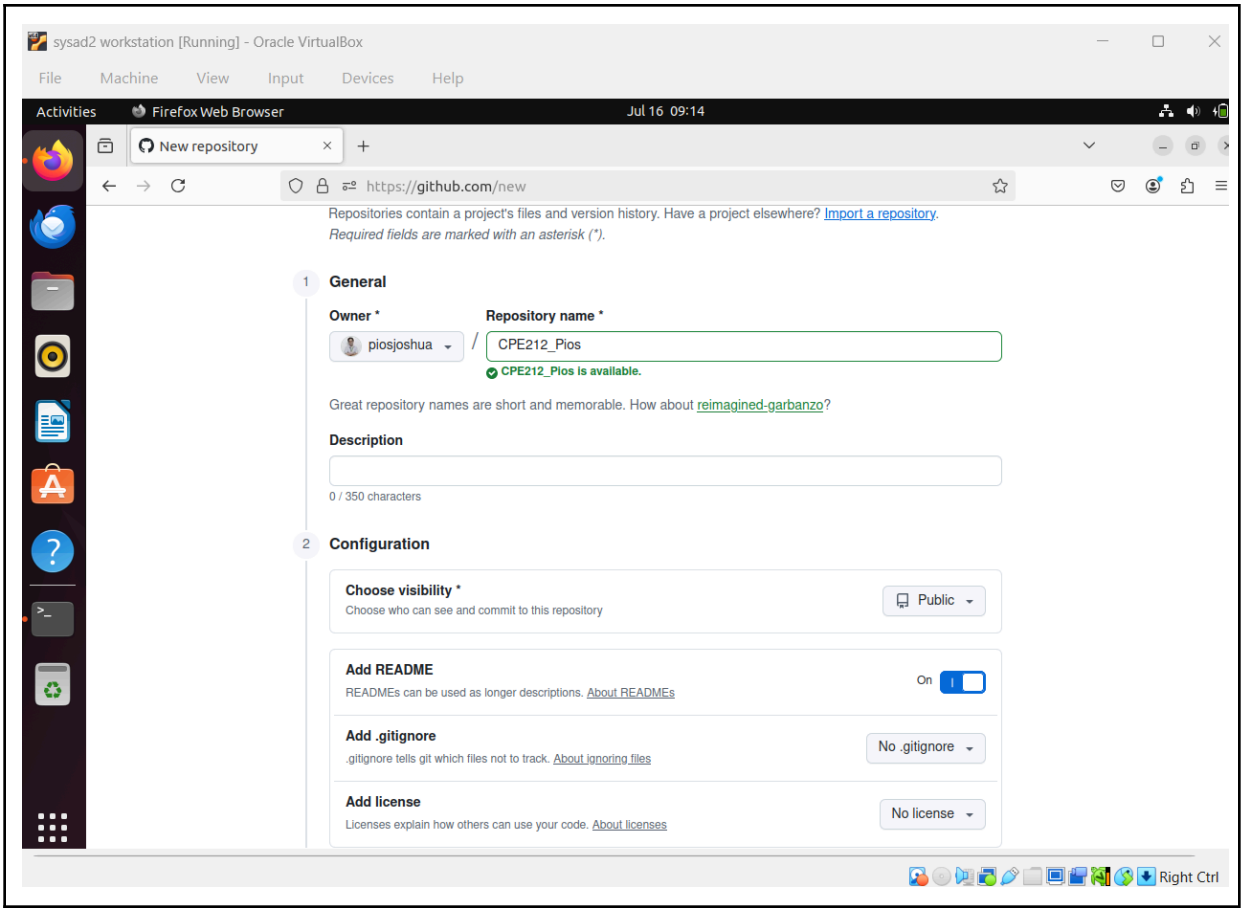
Number of key(s) added: 1

Now try logging into the machine, with:   "ssh 'jpbpios@server2'"
and check to make sure that only the key(s) you wanted were added.

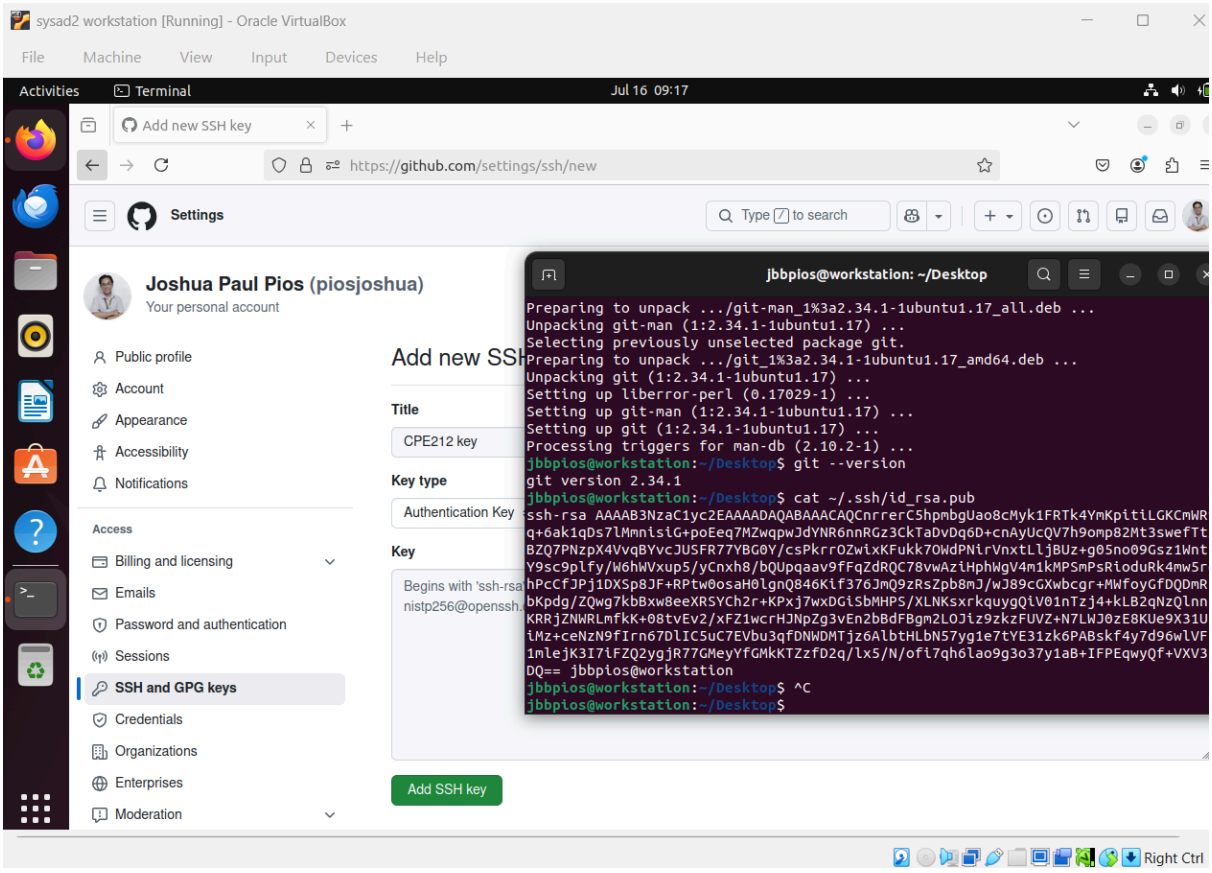
jbbpios@workstation:~/Desktop$
```

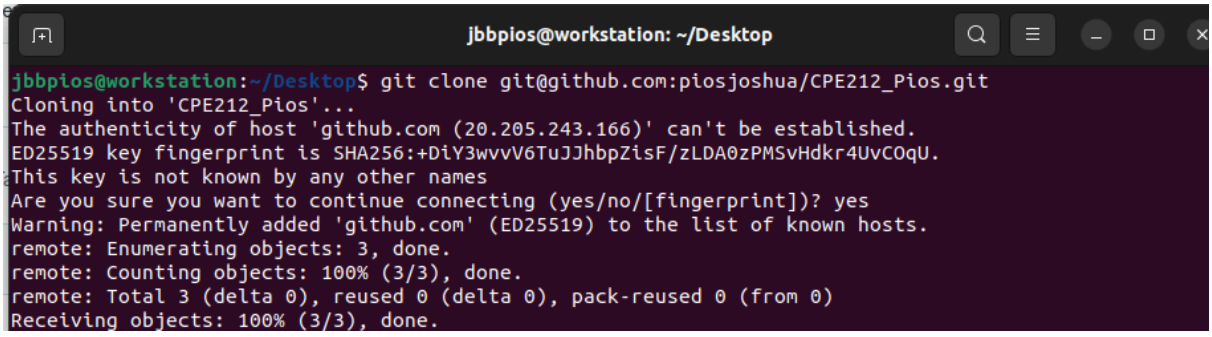
Task 3

```
jbbpios@workstation:~/Desktop$ sudo apt install git
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  git-man liberror-perl
Suggested packages:
  git-daemon-run | git-daemon-sysvinit git-doc git-email git-gui gitk gitweb
  git-cvs git-mediawiki git-svn
The following NEW packages will be installed:
  git git-man liberror-perl
0 upgraded, 3 newly installed, 0 to remove and 439 not upgraded.
Need to get 4,154 kB of archives.
After this operation, 21.0 MB of additional disk space will be used.
Do you want to continue? [Y/n]
0% [Working]
```


sysad2 workstation [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Activities
Firefox Web Browser
Jul 16 09:14
New repository
https://github.com/new
Repositories contain a project's files and version history. Have a project elsewhere? Import a repository.
Required fields are marked with an asterisk (*).
1 General
Owner *
Repository name *
piosjoshua
CPE212_Pios
CPE212_Pios is available.
Great repository names are short and memorable. How about reimagined-garbanzo?
Description
0 / 350 characters
2 Configuration
Choose visibility *
Choose who can see and commit to this repository
Public
Add README
READMEs can be used as longer descriptions. About READMEs
On
Add .gitignore
.gitignore tells git which files not to track. About ignoring files
No .gitignore
Add license
Licenses explain how others can use your code. About licenses
No license
Right Ctrl

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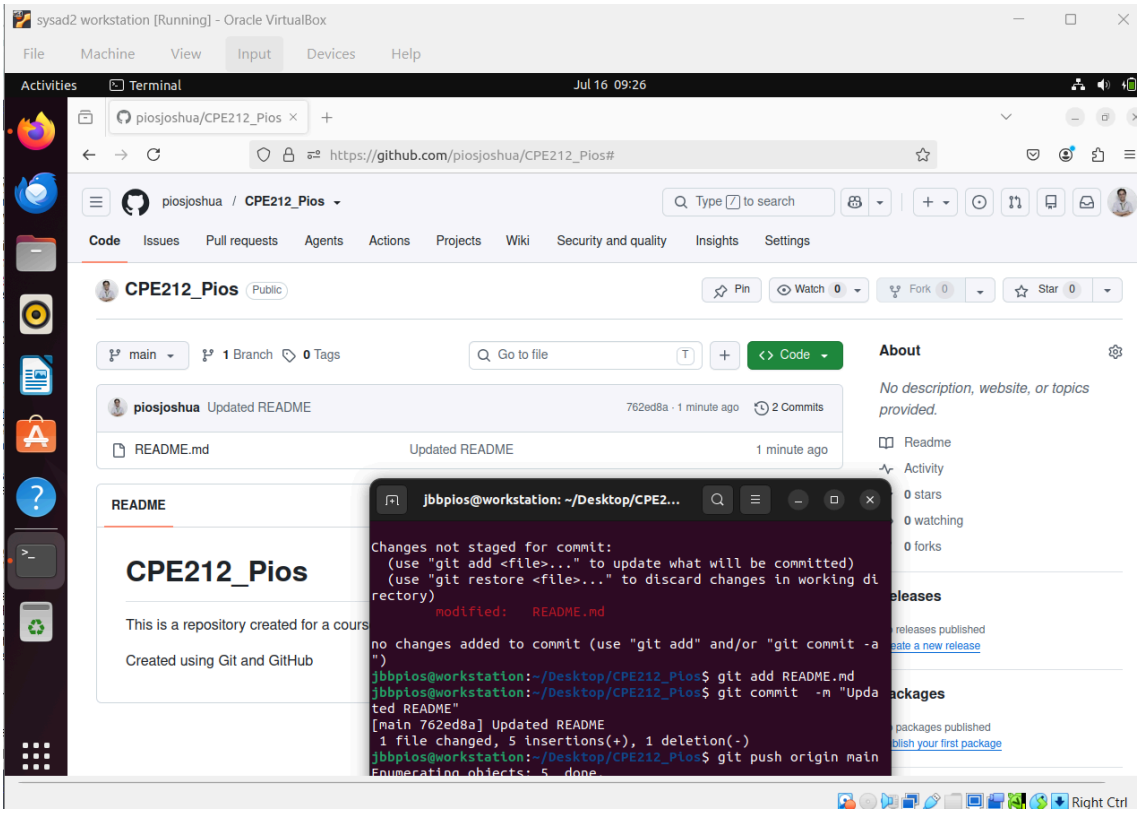




```
jbbpios@workstation:~/Desktop/CPE212_Pios$ nano README.md
jbbpios@workstation:~/Desktop/CPE212_Pios$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   README.md

no changes added to commit (use "git add" and/or "git commit -a")
jbbpios@workstation:~/Desktop/CPE212_Pios$
```



2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Answer the following:

1. How will you describe the ssh-program? What does it do?
   - SSH (Secure Shell) is a secure network protocol that allows users to remotely access and manage another computer over an encrypted connection. It protects data and login credentials while enabling secure file transfers and remote command execution.

2. How do you know that you already installed the public key to the remote servers?
   - I confirmed that the public key was successfully installed because I was able to log in to the remote server using SSH without being prompted for a password. This indicates that the server recognized my public key and authenticated me using the corresponding private key.

3. What sort of things have we so far done to the remote servers using ansible commands?
   - Using Ansible, we automated administrative tasks on the remote servers, such as executing commands, managing configurations, and deploying changes from a central workstation. This made server management faster, more consistent, and reduced the need to log in to each server manually.

4. How important is the inventory file?
   - The inventory file is essential because it tells Ansible which remote servers to manage and how to connect to them. It allows administrators to organize hosts into groups and efficiently run commands or playbooks on one or multiple servers at the same time.

Conclusion/Learnings:

This activity helped me understand how SSH key-based authentication provides a more secure and convenient way to access remote servers without repeatedly entering passwords. I also learned how to set up and use Git with GitHub to manage source code, track changes, and synchronize a local repository with a remote repository. Overall, the activity improved my skills in Linux system administration, secure remote access, version control, and basic server management.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: **N/A**

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: **N/A**

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: **N/A**

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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